Seiberg-Witten Theory and the Intersection Form

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Introduction

FOUR-MANIFOLDS live in a strange world. Very much like three-manifolds, their dimension is high enough so that the classical tools used to study curves and surfaces fall short. On the other hand, in the smooth realm, their dimension is too low, and there is not enough room to apply advanced techniques of differential topology to them.

In the early 1980s, Freedman proved that a fundamental tool for studying manifolds of dimension five and greater could still be applied to four-manifolds if one ignores the smooth structure. Donaldson theory later showed that the smooth world behaves very differently, and Seiberg-Witten theory provided another route into the topology of smooth four-manifolds.

This document is not a reproduction of the original thesis text. It is a compact layout reconstruction using short demonstration text, selected terminology, and the same high-level chapter structure in order to test typography, chapter openings, theorem boxes, drop caps, headers, and the table of contents.

Acknowledgements

First of all, I would like to thank the mathematical community whose documents make LaTeX typography worth studying. This section is included to test the visual spacing of front matter pages.

Chapter 1

Preliminaries

In this chapter we review the basic language of algebraic topology and differential geometry that is needed to discuss Seiberg-Witten theory and the Donaldson theorem. In the differential realm, we assume familiarity with smooth manifolds, de Rham cohomology, vector bundles, and some Riemannian geometry. In algebraic topology, we assume familiarity with singular and cellular homology.

1.1 Gauge theory

Gauge theory is the heart of Seiberg-Witten theory. The name comes from physics, where gauge fields are fields with symmetries and redundant degrees of freedom. Mathematically, the study of gauge fields is done in terms of principal bundles, connections, curvature, and associated vector bundles.

1.1.1 Principal bundles, connections and curvature

The basic objects of gauge theory are principal bundles.

Definition. [Principal G-bundle] Let M be a smooth manifold and let G be a Lie group. A principal G-bundle over M is a smooth manifold P together with a smooth surjective map $\pi\colon P \to M$ and a right action of G on P satisfying equivariance and local triviality conditions.

We write this compactly as

$$G \hookrightarrow P \xrightarrow{\pi} M.$$

Example. [Frame bundles] Let $E \to M$ be a $\mathbb{K}$-vector bundle of rank k. The frame bundle $\mathrm{Fr}(E)$ is the bundle whose fiber over $x \in M$ consists of all ordered bases of E_x. It is naturally a principal $\mathrm{GL}(k, \mathbb{K})$-bundle.

$$\mathrm{Fr}(E) = \coprod_{x \in M} \mathrm{Fr}_x(E).$$

If E is a real Riemannian vector bundle, one obtains the orthogonal frame bundle $O(E)$. If E is a complex Hermitian bundle, one obtains the unitary frame bundle $U(E)$.

Definition. [Connection on a principal bundle] Let G be a Lie group with Lie algebra $\mathfrak{g}$. A connection on a principal bundle $G \hookrightarrow P \to M$ is a $\mathfrak{g}$-valued one-form $\omega \in \Omega^1(P, \mathfrak{g})$ satisfying the standard equivariance and normalization conditions.

Proposition. [Transformation of local gauge potentials] Let ω be a connection on a principal bundle $G \hookrightarrow P \to M$, and let $\{U_i\}_{i \in I}$ be a trivializing cover with local gauges. The local gauge potentials are related on overlaps by

$$\mathcal{A}_j = \mathrm{Ad}_{g_{ij}^{-1}} \circ \mathcal{A}_i + g_{ij}^* \Theta.$$

If G is a matrix Lie group, this becomes

$$\mathcal{A}_j = g_{ij}^{-1} \mathcal{A}_i g_{ij} + g_{ij}^{-1} \mathrm{d} g_{ij}.$$

Definition. [Curvature] The curvature of a connection ω is the two-form

$$\Omega := \mathrm{d}^\omega \omega.$$

Equivalently, it satisfies Cartan's structure equation

$$\Omega = \mathrm{d}\omega + \frac{1}{2}[\omega, \omega].$$

1.1.2 Associated bundles and matter fields

Let $G \hookrightarrow P \to M$ be a principal bundle, V a vector space, and $\rho\colon G \to \mathrm{GL}(V)$ a representation. The associated bundle is

$$P \times_\rho V = P \times V / G.$$

Sections of this bundle can be interpreted as matter fields in the language of physics.

1.1.3 The space of connections

Proposition. [Connections form an affine space] Let $\mathrm{Conn}(P)$ be the set of connections of a principal bundle $G \hookrightarrow P \to M$. Then $\mathrm{Conn}(P)$ is an affine space modeled on the vector space of Ad-tensorial one-forms $\Omega^1_{\mathrm{Ad}}(P, \mathfrak{g})$.

1.2 Algebraic topology

The main algebraic-topological tools needed later are cohomology, characteristic classes, orientations, and the intersection form of four-manifolds.

1.2.1 Cohomology and cups

Definition. [Pairing of cohomology and homology] The natural pairing of cohomology and homology is

$$\langle -, - \rangle \colon H^k(C, G) \otimes H_k(C) \to G,$$

given by evaluating a cohomology class on a homology class.

Theorem. [Universal coefficients for cohomology] Let $C_\bullet$ be a chain complex of free abelian groups and let G be an abelian group. Then there is a split exact sequence

$$0 \to \mathrm{Ext}(H_{k-1}(C), G) \to H^k(C, G) \to \mathrm{Hom}(H_k(C), G) \to 0.$$

1.2.2 Orientations and caps

A local orientation at a point of an n-manifold is a choice of generator of the local homology group. A global orientation is a locally consistent family of such choices.

1.2.3 The intersection pairing

The intersection pairing on a closed oriented four-manifold is one of the central objects in this story. It may be defined using Poincaré duality and the cup product.

1.3 Characteristic classes

Characteristic classes measure obstructions to trivializing bundles and will later be used to formulate the existence of spin and Spin^c structures.

1.3.1 Classifying spaces for line bundles and the first Chern class

Complex line bundles are classified by maps into $\mathbb{C}\mathrm{P}^\infty$, and the first Chern class records the corresponding cohomology class.

1.3.2 Čech cohomology and the first Stiefel-Whitney class

The first Stiefel-Whitney class detects the obstruction to orientability.

1.4 Hodge theory

This section would collect the Hodge-theoretic facts used later: harmonic representatives, Hodge decomposition, and self-dual forms.

1.5 Elliptic regularity

1.5.1 Sobolev spaces

Sobolev spaces provide the functional-analytic setting for elliptic differential operators.

1.5.2 Elliptic operators

Elliptic regularity is one of the analytic mechanisms behind smoothness statements for moduli spaces.

Chapter 2

Spin Geometry

THE PURPOSE of this chapter is to introduce the algebraic and geometric objects that appear in the Seiberg-Witten equations. Clifford algebras, spin groups, spinor bundles, and Dirac operators provide the language in which the equations are formulated.

2.1 Clifford algebras

2.1.1 Definition and basic properties

Definition. [Clifford algebra] Let (V, q) be a quadratic vector space. The Clifford algebra $\mathrm{Cl}(V, q)$ is the quotient of the tensor algebra $T(V)$ by the ideal generated by elements of the form

$$v \otimes v + q(v)1.$$

Notation. We denote by $\mathbb{R}^{r,s}$ the semi-Riemannian vector space $\mathbb{R}^{r+s}$ with r positive and s negative directions. We write $\mathrm{Cl}(r, s)$ for the corresponding Clifford algebra.

Example. [Cl(1) and Cl(2)] For $n = 1$, the real Clifford algebra generated by one element with relation $e^2 = -1$ is isomorphic to $\mathbb{C}$. For $n = 2$, one obtains an algebra isomorphic to the quaternions $\mathbb{H}$.

2.1.2 Complex(ified) Clifford algebras

Complex Clifford algebras are often simpler to classify and are central for describing spin representations.

2.2 The Spin group

2.2.1 Definition and examples

Definition. [Spin group] The group $\mathrm{Spin}(n)$ is the subgroup of the Clifford algebra generated by products of unit vectors and fitting into the double cover

$$1 \to \mathbb{Z}_2 \to \mathrm{Spin}(n) \to \mathrm{SO}(n) \to 1.$$

2.2.2 When $n = 4$

Dimension four is special because $\mathrm{Spin}(4)$ decomposes in a way that makes four-dimensional gauge theory particularly rich.

2.2.3 Lie algebra structures

Lie algebra computations identify infinitesimal versions of the spin actions used later.

2.3 Representations

2.3.1 Classification and examples

Theorem. [Irreducible representations of matrix algebras] Let A be a finite direct sum of matrix algebras over a field. The irreducible representations of A are the projections onto the simple summands, followed by the standard representation.

2.3.2 The Spin representation

Spin representations arise by restricting Clifford algebra representations to the spin group.

2.3.3 Clifford multiplication and infinitesimal actions

Clifford multiplication connects algebraic representations with first-order differential operators.

2.4 Spin structures

2.4.1 From local to global

Definition. [Spin structure] Let M be an oriented Riemannian manifold. A spin structure is a principal Spin(n)-bundle lifting the oriented orthonormal frame bundle of M.

2.4.2 Spinor bundles

A spin structure allows one to construct spinor bundles, whose sections are spinor fields.

2.5 The Dirac operator

2.5.1 Spin connections

Connections on spinor bundles are induced by Levi-Civita connections and auxiliary choices.

2.5.2 The Dirac operator

Definition. [Dirac operator] Given a spinor bundle $S \to M$ with connection ∇, the Dirac operator is locally given by

$$\not{D}\psi = \sum_i e_i \cdot \nabla_{e_i}\psi,$$

where $\{e_i\}$ is a local orthonormal frame.

2.6 The world of Spin^c

2.6.1 The Spin^c group

Definition. [Spin^c group] The group Spin$^c(n)$ is defined by

$$\mathrm{Spin}^c(n) = \frac{\mathrm{Spin}(n) \times U(1)}{\{\pm(1,1)\}}.$$

2.6.2 Going global: Spin^c-structures and spinors

Spin^c structures are weaker than spin structures and exist on a broader class of four-manifolds.

2.6.3 The complex spin connection and the coupled Dirac operator

The coupled Dirac operator is the analytic object appearing in the Seiberg-Witten equations.

2.7 Final ingredient: The squaring map

2.7.1 The linear squaring map

The squaring map sends spinors to self-dual two-forms in a way compatible with the spin representation.

2.7.2 The global squaring map

Globally, this construction produces the nonlinear term in the Seiberg-Witten equations.

Chapter 3

The Seiberg-Witten Equations and Moduli Space

We are finally ready to define the Seiberg-Witten equations and their moduli space. Let M be a closed oriented Riemannian four-manifold with a fixed Spin^c structure. The equations are written for a spinor and a compatible connection.

3.1 The gauge group and its action

Definition. [Gauge group] For a principal $U(1)$-bundle $P \to M$, the gauge group is

$$\mathcal{G} := \mathrm{Aut}_{U(1)}(P).$$

In many settings this may be identified with $C^\infty(M, U(1))$.

3.2 Topology of the Moduli Space

Definition. [Seiberg-Witten equations] The Seiberg-Witten equations for a spinor ψ and a connection A are

$$\not{D}_A \psi = 0, \qquad F_A^+ = \sigma^+(\psi).$$

Definition. [Moduli space] The Seiberg-Witten moduli space is the quotient

$$\mathcal{M} = \{(\psi, A) \mid \not{D}_A \psi = 0,\ F_A^+ = \sigma^+(\psi)\}/\mathcal{G}.$$

3.2.1 Dimension of $\mathcal{M}$

The expected dimension is computed from an elliptic deformation complex and an index theorem.

3.2.2 Generic smoothness

For generic perturbations, the moduli space is smooth under suitable hypotheses.

3.2.3 Orientation

Orienting the moduli space requires orienting a determinant line associated to the deformation complex.

3.2.4 Compactness

Compactness follows from a priori estimates and elliptic regularity.

3.3 The Seiberg-Witten invariant

When the moduli space has the correct dimension, one can extract invariants of the smooth structure of the underlying four-manifold.

Chapter 4

The Intersection Form and Donaldson's Theorem

THE INTERSECTION form is one of the most important algebraic invariants of a closed oriented four-manifold. It connects embedded surfaces, cup products, Poincaré duality, and the classification of manifolds.

4.1 The intersection form of 4-manifolds

Definition. [Intersection form] Let M be a closed, connected, oriented four-manifold. The intersection form is the symmetric bilinear pairing

$$q_M \colon H_2(M, \mathbb{Z}) \times H_2(M, \mathbb{Z}) \to \mathbb{Z}$$

defined by

$$q_M(\sigma, \eta) = \langle \mathrm{PD}(\sigma) \smile \mathrm{PD}(\eta), [M] \rangle.$$

Proposition. If $\alpha, \beta \in H_2(M, \mathbb{Z})$ are represented by embedded surfaces $S_\alpha, S_\beta \subset M$, then

$$q_M(\alpha, \beta) = S_\alpha \cdot S_\beta.$$

4.1.1 Unimodular symmetric forms

Unimodular symmetric forms may be definite or indefinite, even or odd. Their arithmetic properties determine much of the topology of simply connected four-manifolds.

4.1.2 The topology of four-manifolds

Theorem. [Freedman] Every unimodular symmetric bilinear form occurs as the intersection form of a closed, oriented, simply connected topological four-manifold, with the usual uniqueness distinction between even and odd forms.

4.2 Finale: Donaldson's theorem

Theorem. [Donaldson] Let M be a smooth, closed, oriented, simply connected four-manifold. If the intersection form of M is definite, then it is diagonalizable over $\mathbb{Z}$.

This theorem demonstrates that the smooth category is much more rigid than the topological category. Seiberg-Witten theory provides a powerful route to constraints of this kind.